

NEUROSCI 366

Lecture 2

向量、矩阵与概率基础 / Vectors, Matrices, and Probability

Source-aligned · Static · Searchable · Printable

Source files

- Lecture2-FC_NEUROSCI_366_F2025.pdf
- Lecture2-FC_Solutions_NEUROSCI_366_F2025.pdf
- Lecture2-FC_Solutions_NEUROSCI_366_F2025.m

Production version: 2026-08-22

Reader profile: neuroscience undergraduate education; mathematics scaffolded from high-school algebra/trigonometry to the formal computational-neuroscience level.

This companion preserves source-page order, distinguishes source material from necessary scaffolding and extensions, and places hints/answers after the lesson. It is a static PDF: no forms, JavaScript, hidden-answer widgets, passwords, or DRM.

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1 How to use this companion

1. 先用 5-minute diagnostic 暴露前置缺口；不要立即翻答案。
2. 按原笔记页序阅读 source-page units：先看原页，再读 clean reconstruction 与补全逻辑。
3. 遇到公式按“问题 → 符号/shape/units → assumptions → 一行一步推导 → intuition → biological meaning → sanity check → failure”阅读。
4. 完成 Cumulative Knowledge Check；只在需要时依次打开 Hint 1、2、3。
5. 至少隔一个 page break 后再核对 Answer Key，并记录错误类型：concept、symbol、algebra、assumption、correlation/causation。

Source hierarchy. 原笔记是内容权威；本书中的“【必要前置】/【补全推导】/【跨课连接】/【延伸】”是为了补齐数学或迁移，不替代源材料。无法确认的内容必须进入 Uncertainty Log。

2 Learning objectives and prerequisite dependency map

- 定义并区分：对象与 shape。
- 从第一原则推导：线性代数运算。
- 解释几何/计算直觉：概率分布。
- 连接生物学解释：联合/边缘/条件。
- 诊断 assumptions 与 failure modes：Bayes 更新。

Dependency map

对象与 shape → 线性代数运算 → 概率分布 → 联合/边缘/条件 → Bayes 更新

Core question

如何用 shape reasoning 保证线性代数运算合法，并用概率分布把神经数据中的不确定性写成可计算对象？

3 Five-minute prerequisite diagnostic

1. A 是 4×2 ， v 是 2×1 ； Av 的 shape 是什么？（卡住时先看 Source-page unit 1，再看补全推导。）
2. 为什么 $u+A$ 在抽象数学中未定义，但 MATLAB 可能给出结果？（卡住时先看 Source-page unit 2，再看补全推导。）
3. 计算 $u^T v$ ，其中 $u=(0,1)^T$ ， $v=(2,5)^T$ 。（卡住时先看 Source-page unit 3，再看补全推导。）
4. $C=\begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$ 为什么不可逆？（卡住时先看 Source-page unit 4，再看补全推导。）
5. Bernoulli(p) 的 variance 在什么 p 最大？（卡住时先看 Source-page unit 4，再看补全推导。）

Scoring guide: 4–5 confident answers → proceed; 2–3 → use the prerequisite labels; 0–1 → read the formula/notation sheet once before the source-aligned lesson.

4 Source-aligned lesson / 按原笔记页序

4.1 Lecture2-FC_NEUROSCI_366_F2025.pdf • p. 1

Flipped Classroom Exercise Set
NEUROSCI 366, Brain Function Through the Lens of Computation
Discussed: Friday, September 19, 2025

1 Operations with vectors and matrices

Computational neuroscience requires familiarity with vectors and matrices. These mathematical objects are typically introduced in high school algebra and studied deeply in a college linear algebra course. This problem serves to remind students of these fundamental concepts and demonstrate the corresponding MATLAB commands. NOTE: Wikipedia articles and MATLAB help pages are great resources for mathematics and programming.

Consider the following scalars, vectors, and matrices,

$$s = 2, u = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, v = \begin{pmatrix} 2 \\ 5 \end{pmatrix}, w = \begin{pmatrix} 0 \\ 1 \\ 2 \\ 5 \end{pmatrix}, A = \begin{pmatrix} 0 & 0.5 \\ 10 & 1 \\ 2 & 9 \\ 2 & 7 \end{pmatrix}, B = \begin{pmatrix} 4 & 4 \\ 8 & 9 \end{pmatrix}, C = \begin{pmatrix} 3 & 4 \\ 6 & 8 \end{pmatrix}. \quad (1)$$

- What is the general rule from mathematics that determines when scalars, vectors, and/or matrices can be summed with each other? Which pairs of mathematical objects in Eq. 1 can be summed? Evaluate these sums with MATLAB. Warning - MATLAB will not complain if you try to sum s with any of the other variables. How does MATLAB interpret $s + A$, and does this operation correspond to the abstract sum of a scalar and a matrix? What does it correspond to?

NOTE: In this and subsequent bullets, please interpret “pairs” to mean two different mathematical objects. For instance, you should consider pairing u with A but not with itself.

- Which pairs of vectors in Eq. 1 can be combined through an inner product? Evaluate the result by hand and in MATLAB.
- What is geometrical meaning of a vector's norm? How does the norm of a vector relate to the inner product? Evaluate the norm of w , denoted $\|w\|$, with MATLAB.
- Which pairs of vectors in Eq. 1 can be combined through an outer product? How many distinct mathematical objects can be produced by taking outer products between these pairs of vectors?
- What is the general rule from mathematics that determines when a matrix can multiply a vector? What is the dimensionality of the resulting vector? Which pairs of matrices and vectors in Eq. 1 can be combined through a matrix product? Choose one such pair and evaluate the matrix product by hand and in MATLAB.
- What is the general rule from mathematics that determines when two matrices can be multiplied together? What is the size of the resulting matrix? Which pairs of matrices in Eq. 1 can be combined through a matrix product? Choose one such pair and evaluate the matrix product by hand and in MATLAB.

1

Original-note anchor. [Source: Lecture2-FC_NEUROSCI_366_F2025.pdf, p. 1]

Clean reconstruction. Exercise Page 1: scalar/vector/matrix objects; 维度规则; sum、inner product、norm、outer product、matrix-vector 与 matrix-matrix multiplication.

What the note is saying. 本页的中心对象是对象与 shape 与线性代数运算。原笔记将它们放进以下链条: scalar/vector/matrix objects; 维度规则; sum、inner product、norm、outer product、matrix-vector 与 matrix-matrix multiplication.

Why it follows. 完整覆盖 dot product、norm、outer product、identity、transpose、inverse、determinant、singular matrix、Hadamard product, 以及 `*`、`.*`、`'`、`inv`、`det`、`eye`。其逻辑后果是: 完整讲 marginalization、independence、conditional probability、product rule 与 Bayes rule; 严格区分 probability、likelihood、prior、posterior。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions.

Figure reading. 把图与矩阵 shape 对齐: 轴代表变量或 mode, 椭圆方向对应 eigenvectors, 轴长/曲率对应 eigenvalues; 不要把统计轴自动解释为因果通路。

Stop & Predict

A 是 4×2 , v 是 2×1 ; Av 的 shape 是什么?

4.2 Lecture2-FC_NEUROSCI_366_F2025.pdf • p. 2

- Explain the concept of an identity matrix. Write down an example of one, and generate it in MATLAB. Note that an identity matrix is usually denoted as I .
- What is the definition of the matrix transpose? Evaluate the transpose of A , denoted A^T , in MATLAB. Which vectors in Eq. 1 can A^T multiply?
- By interpreting an N -dimensional vector as an $N \times 1$ matrix, the transpose of a vector can be interpreted as a $1 \times N$ matrix, termed a row vector. Write the norm of w in terms of w and w^T .
- What is the definition of the matrix inverse? Explain why A is not invertible. Evaluate the inverse of B , denoted B^{-1} , in MATLAB. Note that C is not invertible because the second row is a scalar multiple of the first, and a matrix is only invertible when its rows and columns are linearly independent. What happens in you try to evaluate its inverse in MATLAB?
- For which mathematical objects in Eq. 1 is the determinant mathematically defined? Evaluate these determinants by hand and in MATLAB. What condition on the determinant determines whether a matrix is invertible?
- The Hadamard product is a mathematical operation that combines same-size vectors or matrices by component-wise multiplication. It has little significance to pure mathematics, but it is extremely useful in programming. Which pairs of mathematical objects in Eq. 1 can be combined with a Hadamard product? Evaluate these Hadamard products with MATLAB.

2 Random processes and probability

2.1 Discrete variables - Bernoulli distribution

The Bernoulli distribution describes a simple binary random process that takes the value 1 with probability p and the value 0 with probability $q = 1 - p$. It is called a discrete probability distribution because it can only take values from a discrete set of outcomes (i.e., $\{0, 1\}$). For instance, you can think of this random process as a biased coin flip where “heads” corresponds to the value 1 and “tails” corresponds to the value 0. What is the probability mass function and cumulative density function of the Bernoulli distribution? Use the probability mass function to derive the mean and variance of the Bernoulli distribution as a function of p . What is the standard deviation of the Bernoulli distribution?

2.2 Continuous variables. Exponential distribution

Some probabilistic processes take continuous values. For example, the exponential distribution assigns non-zero probability to any non-negative value according to the probability density function

$$P(x) = \lambda e^{-\lambda x} \quad (2)$$

2

Original-note anchor. [Source: Lecture2-FC_NEUROSCI_366_F2025.pdf, p. 2]

Clean reconstruction. Exercise Page 2: identity、transpose、vector-as-matrix、inverse、singularity、determinant、Hadamard product; Bernoulli distribution; exponential distribution 开始。

What the note is saying. 本页的中心对象是概率分布与联合/边缘/条件。原笔记将它们放进以下链条：identity、transpose、vector-as-matrix、inverse、singularity、determinant、Hadamard product; Bernoulli distribution; exponential distribution 开始。

Why it follows. 完整覆盖 dot product、norm、outer product、identity、transpose、inverse、determinant、singular matrix、Hadamard product, 以及 `*`、`.*`、`'`、`inv`、`det`、`eye`。其逻辑后果是：完整推导 Bernoulli mean/variance; exponential PDF、CDF、mean、median、mode、standard deviation, 并给出 integration by parts 的最小支架。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 把图与矩阵 shape 对齐：轴代表变量或 mode，椭圆方向对应 eigenvectors，轴长/速率对应 eigenvalues；不要把统计轴自动解释为因果通路。

Stop & Predict

为什么 $u+A$ 在抽象数学中未定义，但 MATLAB 可能给出结果？

4.3 Lecture2-FC_NEUROSCI_366_F2025.pdf • p. 3

for $x \geq 0$, where λ is a positive parameter called the rate. Evaluate the cumulative density function. What is the probability that x takes a value in the interval (a, b) for $0 \leq a \leq b$? Calculate the mean, median, mode, and standard deviation of the exponential distribution.

3 Multivariate probability distributions

Multivariate probability distributions allow one to consider the joint outcome of multiple noisy measurements. For instance, a neuroscientist might measure the number of action potentials emitted by many different neurons on the same trial.

3.1 Bivariate normal distribution

The multivariate normal distribution is a very important example that will allow us to illustrate many fundamental concepts. Specializing to the case of two continuous variables, x_1 and x_2 , the probability density function of the bivariate normal distribution is

$$P(x_1, x_2) = P(x) = \frac{1}{\sqrt{(2\pi)^2 \det C}} e^{-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)}, \quad (3)$$

where we've combined x_1 and x_2 into a 2-dimensional vector $x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$, $\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}$ is a 2-dimensional vector of means, C is a 2×2 covariance matrix, $C = \begin{pmatrix} \sigma_1^2 & c\sigma_1\sigma_2 \\ c\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}$, and $\det C$ is the determinant of C . The bivariate normal distribution has five parameters, μ_1 , μ_2 , σ_1 , σ_2 , and c ? Explain the meaning of each of these parameters.

3.2 Marginal distributions

Without doing a new calculation, what do you think will be the probability density function for x_1 alone if x_1 and x_2 are bivariate normal? In mathematical terms, this is called the marginal distribution of x_1 . It can generally be computed from the joint distribution via $P(x_1) = \int_{-\infty}^{\infty} dx_2 P(x_1, x_2)$.

3.3 Statistical independence

We say that x_1 and x_2 are statistically independent if $P(x_1, x_2) = P(x_1)P(x_2)$. Assuming that x_1 and x_2 are bivariate normal, when are they statistically independent?

3.4 Conditional distributions

It is often useful to compute the probability distribution of one variable under the condition that the value of another has some known value. This is called a conditional probability distribution, and it can be computed as $P(x_1|x_2) = \frac{P(x_1, x_2)}{P(x_2)}$. In general, what is the conditional probability distribution of x_1 on x_2 if x_1 and x_2 are statistically independent? Consider a bivariate normal distribution with $\mu_1 = 0$, $\mu_2 = 1$, $\sigma_1 = 0.5$, $\sigma_2 = 1$, and $c = 0.7$. Is the mean of $P(x_1|x_2 = 0.75)$ larger or smaller than the mean of $P(x_1)$. Why?

3

Original-note anchor. [Source: Lecture2-FC_NEUROSCI_366_F2025.pdf, p. 3]

Clean reconstruction. Exercise Page 3: exponential CDF/mean/median/mode/standard deviation; bivariate normal; covariance matrix; marginal, independence, conditional distribution. **What the note is saying.** 本页的中心对象是 Bayes 更新与对象与 shape。原笔记将它们放进以下链条: exponential CDF/mean/median/mode/standard deviation; bivariate normal; covariance matrix; marginal, independence, conditional distribution.

Why it follows. 逐项解释 bivariate Gaussian 的 mean vector、covariance matrix、determinant、inverse、correlation coefficient; 用图形解释 covariance 的符号。其逻辑后果是: 完整推导 Bernoulli mean/variance; exponential PDF、CDF、mean、median、mode、standard deviation, 并给出 integration by parts 的最小支架。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions.

Figure reading. 先明确横纵轴、单位、归一化与每条曲线改变的唯一参数; 区分示意趋势、模型预测和真正测得的数据点。

Stop & Predict

计算 $u^T v$, 其中 $u = (0, 1)^T$, $v = (2, 5)^T$.

4.4 Lecture2-FC_NEUROSCI_366_F2025.pdf • p. 4

3.5 Bayes rule

Use the definition of the conditional probability distribution to write $P(x_2|x_1)$ in terms of $P(x_1)$, $P(x_2)$, and $P(x_1|x_2)$. This simple equation is known as Bayes rule or Bayes theorem.

4

Original-note anchor. [Source: Lecture2-FC_NEUROSCI_366_F2025.pdf, p. 4]

Clean reconstruction. Exercise Page 4: Bayes rule.

What the note is saying. 本页的中心对象是线性代数运算与概率分布。原笔记将它们放进以下链条：Bayes rule。

Why it follows. 完整讲 marginalization、independence、conditional probability、product rule 与 Bayes rule；严格区分 probability、likelihood、prior、posterior。其逻辑后果是：完整推导 Bernoulli mean/variance；exponential PDF、CDF、mean、median、mode、standard deviation，并给出 integration by parts 的最小支架。后面的正式推导会逐项写出 assumptions、shape/units、limiting cases 与 failure conditions。

Figure reading. 先识别图中对象、箭头、参数和坐标系，再问它表达的是定义、推导、模型预测还是经验观察；示意图不提供未标注的定量精度。

Stop & Predict

C=[[3,4],[6,8]] 为什么不可逆？

5 MATLAB source audit / 代码逐行审计

Authority. The listing below is the actual uploaded source; instructional intent and production-quality recommendations are kept separate.

```

1 %% Clear workspace before starting work
2
3 clear all;
4
5 %% Problem 1
6
7 % Bullet 1
8 s = 2;
9 u = [0; 1];
10 v = [2; 5];
11 w = [0; 1; 2; 5];
12 A = [0 0.5; 10 1; 2 9; 2 7];
13 B = [4 4; 8 9];
14 C = [3 4; 6 8];
15
16 % Well-defined mathematical objects
17 u+v
18 B+C
19
20 % Imprecise MATLAB notation
21 s+A
22 s+A-A
23
24 % Bullet 2
25 dot(u,v)
26
27 % Bullet 3
28 norm(w)
29 sqrt(dot(w,w))
30
31 % Bullet 5
32 A*v
33
34 % Bullet 6
35 B*C
36
37 % Bullet 7
38 eye(3)
39
40 % Bullet 8
41 A'
42
43 % Bullet 10
44 inv(B)
45 inv(C)
46
47 % Bullet 11
48 det(s)
49 det(B)
50 det(C)
51
52 % Bullet 12
53 u.*v
54 B.*C

```

Lines	Role	Audit finding	Consequence
1–3	Workspace reset	<code>clear all</code> 可复现课程环境, 但会清函数缓存; 日常更常用 <code>clearvars; close all; clc</code> .	No mathematical effect, but avoid unnecessary global reset.
8–14	Objects and scalar expansion	<code>s + A</code> relies on MATLAB scalar expansion. This is valid MATLAB, but it is not “matrix addition of unequal shapes.”	Separate language semantics from abstract linear algebra.
17–22	Sums and products	<code>sum(A)</code> reduces by columns by default; <code>u*v</code> is dimensionally invalid while <code>u*v'</code> is an outer product.	Always state output shape.
25–29	Matrix multiplication	<code>A*v</code> , <code>B*u</code> , <code>A*B</code> , <code>B*A</code> are determined by inner dimensions; only defined products should be interpreted.	Predict shape before execution.
32–44	Inverse, determinant, singularity	The script demonstrates explicit inversion for B. Production numerical solving should use left division. C is singular because <code>det(C)=0</code> ; <code>inv(C)</code> is not a mathematical inverse.	Do not use determinant alone as a stable numerical singularity test.
47–54	Hadamard product	<code>A.*A</code> is elementwise and distinct from <code>A*A</code> .	Same symbols, different algebra.

6 补全推导与数学支架 / Derivations

6.1 shape-first 线性代数

【必要前置】 判断加法、内积、外积和矩阵乘法是否定义。

$$A \in \mathbb{R}^{m \times n}, x \in \mathbb{R}^{n \times 1} \Rightarrow Ax \in \mathbb{R}^{m \times 1}$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

$$u^T v \in \mathbb{R}, \quad uv^T \in \mathbb{R}^{m \times n}$$

1. 加法要求 shape 完全相同；矩阵乘法要求内维相等。
2. 内积把两个同维列向量压缩成 scalar；外积保留每一对分量组合形成 matrix。
3. Hadamard product 逐元素相乘，要求同 shape；它与矩阵乘法不是同一运算。
4. 每一步先写 shape，再算数值，可提前排除大多数错误。

Geometric/computational intuition. 矩阵乘法是“每个输出行模板”与输入向量做内积；外积则构造所有成对乘积。

Biological interpretation / failure condition. MATLAB implicit expansion 可能让 shape 不同的对象产生输出，但该输出未必对应题目所问的抽象运算。

Micro-check

$C = \begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$ 为什么不可逆？

6.2 Bernoulli 的 mean 与 variance

【补全推导】 从 PMF 推出二元随机变量的中心与波动。

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

$$E[X] = p$$

$$\text{Var}(X) = E[X^2] - E[X]^2 = p(1 - p)$$

1. X 只能取 0 或 1, 所以 $X^2=X$ 。
2. $E[X]=(1-p) \times 0 + p \times 1 = p$ 。
3. $E[X^2]=E[X]=p$, 因此 $\text{variance}=p-p^2$ 。
4. $p=0$ 或 1 时没有随机性; $p=1/2$ 时 variance 最大。

Geometric/computational intuition. variance 衡量 coin flip 离其平均值的平方距离; 最不确定的硬币波动最大。

Biological interpretation / failure condition. Bernoulli 只描述单次 binary outcome; 多个相关 trial 不能简单当作独立 Bernoulli。

Micro-check

Bernoulli(p) 的 variance 在什么 p 最大?

6.3 Exponential distribution 的 CDF、均值与尺度

【跨页连接】 描述等待时间并从 PDF 推导可观测统计量。

$$f(x) = \lambda e^{-\lambda x} \mathbf{1}_{x \geq 0}$$

$$F(x) = 1 - e^{-\lambda x} \quad (x \geq 0)$$

$$E[X] = \frac{1}{\lambda}, \quad \text{median} = \frac{\ln 2}{\lambda}, \quad \text{SD}(X) = \frac{1}{\lambda}$$

1. CDF 是从负无穷积分到 x；负值区间概率为零。
2. 区间概率由 $F(b) - F(a) = e^{-\lambda a} - e^{-\lambda b}$ 。
3. 均值用 integration by parts；第二矩给出 $2/\lambda^2$ ，故 $\text{variance} = 1/\lambda^2$ 。
4. mode 在 $x=0$ ，因为 density 随 x 单调下降。

Geometric/computational intuition. λ 是 rate；rate 越大，典型等待时间 $1/\lambda$ 越短。

Biological interpretation / failure condition. 若 hazard 随时间改变或事件有 refractory/history dependence，指数等待时间模型不成立。

Micro-check

Exponential(λ) 的 mean 与 SD 分别是多少？

6.4 Bivariate Gaussian、conditional mean 与 Bayes

【模型边界】 把两个有相关性的连续测量写成联合分布并进行条件更新。

$$C = \begin{bmatrix} \sigma_1^2 & c\sigma_1\sigma_2 \\ c\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$$

$$E[X_1 | X_2 = x_2] = \mu_1 + c \frac{\sigma_1}{\sigma_2} (x_2 - \mu_2)$$

$$P(x_2 | x_1) = \frac{P(x_1 | x_2)P(x_2)}{P(x_1)}$$

1. c 的符号决定椭圆倾斜方向，绝对值决定拉长程度。
2. Gaussian 情形下 $c=0$ 等价于 independence；一般分布中 uncorrelated 不保证 independent。
3. 条件均值沿相关方向移动；给定 $x_2 < \mu_2$ 且 $c > 0$ ， x_1 的条件均值低于 μ_1 。
4. Bayes 用 product rule 重排同一 joint probability。

Geometric/computational intuition. conditioning 相当于在联合密度椭圆上切一条 slice，再重新归一化。

Biological interpretation / failure condition. 把 likelihood 当作 parameter 的概率分布、或忘记 evidence 的归一化作用。

Micro-check

为什么连续变量 $P(X=x)=0$ 不与 density $f(x)>0$ 矛盾？

7 Cross-page synthesis / 跨页因果与数学链

本课建立课程后续几乎所有公式的语法：先辨认 scalar、vector、matrix 的 shape，再判断运算是否定义；概率部分则从单变量 Bernoulli/Exponential 走向 bivariate Gaussian、marginal、conditional 与 Bayes。MATLAB 可以执行的表达式不一定等于抽象数学中定义的运算，因此代码输出必须和对象含义分开解释。

Lecture 2 — key reconstruction (schematic)

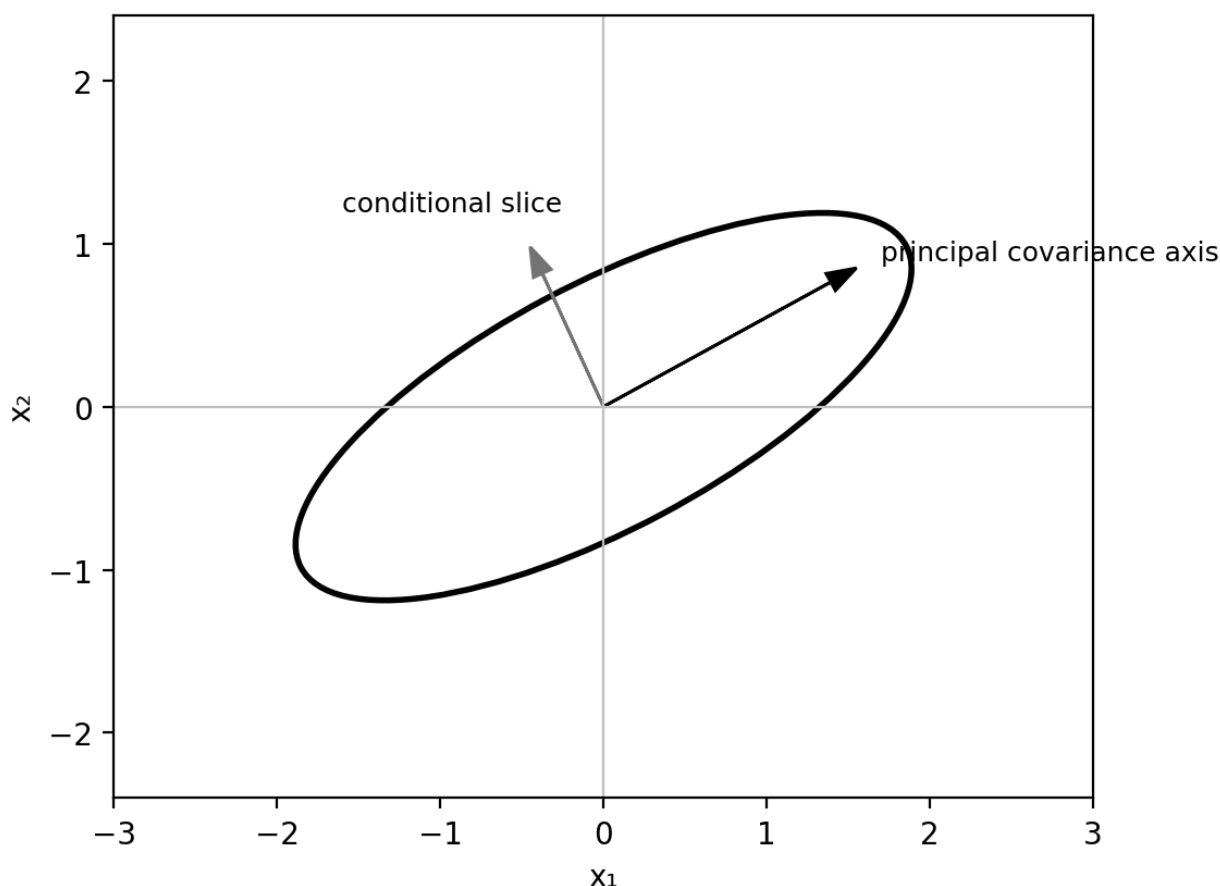


Figure note: schematic reconstruction; it encodes qualitative relations from the lesson and does not invent precise empirical data points.

8 Worked examples and transfer

8.1 题目给定对象的手算

【原课/核心例题】 Prompt. $u=(0,1)^T, v=(2,5)^T, w=(0,1,2,5)^T, A$ 为 $4 \times 2, B=[[4,4],[8,9]], C=[[3,4],[6,8]]$ 。

Worked solution. $u+v=(2,6)^T; u^T v=5; ||w||=\sqrt{30}; Av=(2.5, 25, 49, 39)^T; BC=[[36, 48], [78, 104]];$
 $\det(B)=4, B^{-1}=(1/4)[[9, -4], [-8, 4]];$ $\det(C)=0$ 。

Check. 每个结果先核对 shape，再核对数值；C 的第二行是第一行的 2 倍，因此 singular。

8.2 conditional Gaussian 数值检查

【跨课连接】 Prompt. $\mu_1=0, \mu_2=1, \sigma_1=0.5, \sigma_2=1, c=0.7$, 给定 $x_2=0.75$ 。

Worked solution. 条件均值为 $0+0.7\times 0.5\times (0.75-1)=-0.0875$, 小于边缘均值 0。

Check. 正相关且观测到的 x_2 低于其均值, 所以对 x_1 的更新也向下。

Micro-check

Explain in your own words. 用不超过 120 字解释: 本课从 source page 1 到最后一页, 究竟把哪一个输入、状态、统计量或学习规则变成了可检验 prediction?

Self-reflection. 你最可能犯的是 concept、symbol、algebra、assumption 还是 causality error? 写出一个具体例子。

9 Formula and notation sheet

1. 矩阵-向量乘法的 shape 规则

$$A_{m \times n} x_{n \times 1} \rightarrow y_{m \times 1}$$

Conditions / conventions: 内维相等

2. inner product

$$u^T v = \sum_i u_i v_i$$

Conditions / conventions: 向量同维

3. outer product

$$uv^T$$

Conditions / conventions: 结果 shape 为 $\text{len}(u) \times \text{len}(v)$

4. Euclidean norm

$$\|w\|_2 = \sqrt{w^T w}$$

Conditions / conventions: 列/行 convention 一致

5. variance 恒等式

$$\text{Var}(X) = E[X^2] - E[X]^2$$

Conditions / conventions: 二阶矩存在

6. CDF 定义

$$F_X(x) = P(X \leq x)$$

Conditions / conventions: 离散/连续均适用

7. exponential PDF

$$f_X(x) = \lambda e^{-\lambda x} \mathbf{1}_{x \geq 0}$$

Conditions / conventions: $\lambda > 0$

8. 二维 Gaussian density

$$p(x) = \frac{1}{2\pi \sqrt{\det C}} e^{-\frac{1}{2}(x-\mu)^T C^{-1}(x-\mu)}$$

Conditions / conventions: C 对称正定

9. Bayes rule

$$P(A | B) = \frac{P(B | A)P(A)}{P(B)}$$

Conditions / conventions: $P(B) > 0$

10 Bilingual glossary

中文	English / symbol	Operational meaning
标量	scalar	单个数；可视为 1×1 对象但需说明语境。
向量	vector	有序分量；本课默认列向量。
矩阵	matrix	按行列组织的二维线性映射表示。
内积	inner product	同维向量对应分量乘积之和。
外积	outer product	列向量与行向量相乘得到矩阵。

中文	English / symbol	Operational meaning
Hadamard 乘积	Hadamard product	同 shape 对象逐元素相乘。
奇异矩阵	singular matrix	不可逆、determinant 为零的方阵。
概率质量函数	probability mass function, PMF	离散结果各自的概率。
概率密度函数	probability density function, PDF	连续变量单位区间概率的密度。
边缘分布	marginal distribution	对其他变量求和/积分后保留一个变量。
条件分布	conditional distribution	已知另一事件/变量时的重新归一化分布。
似然	likelihood	把观测固定后，模型参数的相对支持度函数。
先验	prior	观测当前数据前对未知量的分布。
后验	posterior	结合 likelihood 与 prior 后的更新分布。

11 Common traps, assumptions, and limitations

1. 把 MATLAB 的 scalar expansion 误当成抽象线性代数中的 scalar+matrix。
2. 混淆 * 与 .*；前者是矩阵乘法，后者是 Hadamard product。
3. 先算数字再看 shape，导致把未定义运算算成结果。
4. 把 density 值当成某个精确连续数值的概率。
5. 认为 uncorrelated 总是 independent；只有 Gaussian 等特定情形成立。
6. 把 posterior、likelihood 和 prior 的 conditioning 方向写反。

Course-specific risk boundary. 审计.m 中 `clear all`、`inv` 等写法：忠实解释课程代码，同时指出数值计算中通常更偏好线性求解而非显式求逆。将 MATLAB 的 scalar-plus-matrix 行为与抽象线性代数严格分开。

12 Cumulative Knowledge Check

Do not turn to the Hint Bank until you have written a first attempt.

1. A 是 4×2 , v 是 2×1 ; Av 的 shape 是什么?

2. 为什么 $u+A$ 在抽象数学中未定义, 但 MATLAB 可能给出结果?

3. 计算 $u^T v$, 其中 $u=(0,1)^T$, $v=(2,5)^T$.

4. $C=\begin{bmatrix} 3 & 4 \\ 6 & 8 \end{bmatrix}$ 为什么不可逆?

5. $\text{Bernoulli}(p)$ 的 variance 在什么 p 最大?

6. $\text{Exponential}(\lambda)$ 的 mean 与 SD 分别是多少?

7. 为什么连续变量 $P(X=x)=0$ 不与 density $f(x)>0$ 矛盾?

8. 二维 Gaussian 中 $c>0$ 的等密度椭圆如何倾斜?

9. 一般情况下 uncorrelated 是否推出 independent?

10. 写出 Bayes rule 并标出 evidence.

11. 为什么 $\text{inv}(A)*b$ 通常不如 $A \setminus b$?

12. Hadamard product $B.*C$ 的 (1,2) 元素是什么?

13. 公式表中的“矩阵-向量乘法的 shape 规则”解决什么问题? 写出适用条件与一个 sanity check.

14. 公式表中的“inner product”解决什么问题? 写出适用条件与一个 sanity check.

15. 公式表中的“outer product”解决什么问题? 写出适用条件与一个 sanity check.

16. 公式表中的“Euclidean norm”解决什么问题？写出适用条件与一个 sanity check。

17. 公式表中的“variance 恒等式”解决什么问题？写出适用条件与一个 sanity check。

18. 公式表中的“CDF 定义”解决什么问题？写出适用条件与一个 sanity check。

13 Hint Bank

1. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
2. **Hint 1.** 先识别对象、变量与假设。
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12. **Hint 1.** 先识别对象、变量与假设。
Hint 2. 写出最相关的定义或方程，再代入。
Hint 3. 检查单位、shape、符号与极限情况。
13. **Hint 1.** 先从“矩阵-向量乘法的 shape 规则”对应的变量关系入手。
Hint 2. 把条件“内维相等”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。
14. **Hint 1.** 先从“inner product”对应的变量关系入手。
Hint 2. 把条件“向量同维”逐字转换为 assumption。
Hint 3. 写公式，再检查至少一种极端参数或单位。

15. **Hint 1.** 先从 “outer product” 对应的变量关系入手。
Hint 2. 把条件 “结果 shape 为 $\text{len}(u) \times \text{len}(v)$ ” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
16. **Hint 1.** 先从 “Euclidean norm” 对应的变量关系入手。
Hint 2. 把条件 “列/行 convention 一致” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
17. **Hint 1.** 先从 “variance 恒等式” 对应的变量关系入手。
Hint 2. 把条件 “二阶矩存在” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。
18. **Hint 1.** 先从 “CDF 定义” 对应的变量关系入手。
Hint 2. 把条件 “离散/连续均适用” 逐字转换为 assumption。
Hint 3. 写公式, 再检查至少一种极端参数或单位。

14 Complete Answer Key with reasoning

1. 4×1 。
2. shape 不同所以抽象加法未定义；MATLAB 把 scalar/小数组广播到 A 的每个元素。
3. $0 \times 2 + 1 \times 5 = 5$ 。
4. 两行线性相关, $\det(C) = 24 - 24 = 0$ 。
5. $p = 1/2$, 此时 $p(1-p) = 1/4$ 。
6. 二者都是 $1/\lambda$ 。
7. 单点区间宽度为零；概率是 density 在区间上的积分。
8. 沿 x_1 与 x_2 同时增大/减小的正斜率方向拉长。
9. 否；independent 推出 uncorrelated (矩存在), 反向一般不成立。
10. $P(A|B) = P(B|A)P(A)/P(B)$, $P(B)$ 是 evidence/normalizer。
11. 显式逆更慢、数值误差更大；线性求解器利用分解直接求解。
12. $4 \times 4 = 16$ 。
13. 适用条件/约定：内维相等。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$A_{m \times n} x_{n \times 1} \rightarrow y_{m \times 1}$$

14. 适用条件/约定：向量同维。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$u^T v = \sum_i u_i v_i$$

15. 适用条件/约定：结果 shape 为 $\text{len}(u) \times \text{len}(v)$ 。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$uv^T$$

16. 适用条件/约定：列/行 convention 一致。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\|w\|_2 = \sqrt{w^T w}$$

17. 适用条件/约定：二阶矩存在。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$\text{Var}(X) = E[X^2] - E[X]^2$$

18. 适用条件/约定：离散/连续均适用。sanity check 应检查单位、符号、极限或 shape 是否与定义一致。

$$F_X(x) = P(X \leq x)$$

15 Source-page concordance and coverage audit

Source file	Page	Coverage status
Lecture2-FC_NEUROSCI_366_F2025.pdf	1	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture2-FC_NEUROSCI_366_F2025.pdf	2	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture2-FC_NEUROSCI_366_F2025.pdf	3	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture2-FC_NEUROSCI_366_F2025.pdf	4	Main source-page unit: thumbnail, reconstruction, explanation, prediction question.
Lecture2-FC_Solutions_NEUROSCI_366_F2025.pdf	1	Answer/verification source: cross-checked against exercises and incorporated in Answer Key.
Lecture2-FC_Solutions_NEUROSCI_366_F2025.pdf	2	Answer/verification source: cross-checked against exercises and incorporated in Answer Key.
Lecture2-FC_Solutions_NEUROSCI_366_F2025.pdf	3	Answer/verification source: cross-checked against exercises and incorporated in Answer Key.
Lecture2-FC_Solutions_NEUROSCI_366_F2025.pdf	4	Answer/verification source: cross-checked against exercises and incorporated in Answer Key.
Lecture2-FC_Solutions_NEUROSCI_366_F2025.pdf	5	Answer/verification source: cross-checked against exercises and incorporated in Answer Key.
Lecture2-FC_Solutions_NEUROSCI_366_F2025.m	lines 1–55	Full source listing and line-group audit included; syntax, shape, units, indexing, numerical stability and edge cases checked.

16 Errata, uncertainty log, and external endnotes

16.1 Errata / source cautions

1. Solutions p.3 的 Bernoulli variance 推导最后一行左侧写成 μ , 语义上应为 σ^2 。
2. MATLAB `clear all` 会清除函数缓存, 教学脚本通常用 `clearvars` 更合适。
3. `inv(C)` 对 singular matrix 会产生警告和不可靠的 Inf/NaN/巨大数值; 不应把输出解释为数学逆。
4. 数值计算中解 $Bx=y$ 通常使用 `B\y`, 而不是显式 `inv(B)*y`。

16.2 Uncertainty log

No unresolved handwriting uncertainty changes a core definition, equation, figure interpretation, or answer in this companion. Where handwriting was potentially ambiguous, the source-page image is preserved and the reconstruction avoids claiming unverified detail.

16.3 External endnotes

No external web or textbook source was used to replace the uploaded notes. Added material is limited to necessary mathematical scaffolding, explicit assumptions, standard sanity checks, and clearly labeled transfer examples.

16.4 Final self-audit

All source pages listed above were visually inspected. Equations were checked for symbol definitions, units/shapes, assumptions, algebraic transitions, limiting cases, and failure conditions. The PDF is static and contains no interactive form fields or scripts.